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DEPARTMENT OF CHEMISTRY
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

Set-1

INTERNAL ASSESSMENT – I (CLA1-T1)

Program: B.Tech
Course Code & Title:21CYB101J & Chemistry
Year &Sem: I Year & I Sem

Date: 13-09-2023
Duration: 12:30 – 1:20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 The magnetic moment of complex ion is 2.83 BM. The complex ion is:

(a) $[V(H_2O)_6]^{3+}$ (b) $[Cr(H_2O)_6]^{3+}$

(c) $[Cu(CN)_4]^{2-}$ (d) $[Mn(Cl)_4]^{2-}$

2 A coordination complex's core atom/ion is also known as

(a) Bronsted-Lowry acid (b) Lewis base

(c) Lewis acid (d) Bronsted-Lowry base

3 A $[M(H_2O)_6]^{2+}$ complex typically absorbs at around 600 nm. It is allowed to react with ammonia to form a new complex $[M(NH_3)_5]^{2+}$ that should have absorption at:

(a) 800 nm (b) 580 nm (c) 620 nm (d) 320 nm

4 Which of the following compound shows optical isomerism?

(a) cis-$[CrCl_3(NH_3)_3]$

(b) cis-$[Co(NH_3)_4Cl_2]^+$

(c) $[Co(en)_3]^{3+}$

(d) trans-$[Co(en)_2Cl_2]^+$

5 The Crystal Field Stabilization Energy (CFSE) and magnetic moment (spin-only) of an octahedral aqua complex of a metal ion (M^{z+}) are $-0.8\ \Delta_0$ and 3.87 BM, respectively. Identify (M^{z+}) :

(a) V^{3+} (b) Cr^{3+} (c) Mn^{4+} (d) Co^{2+}

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Part – B (2 x 10 = 20 Marks)

6. a. Explain crystal field splitting of d orbitals with neat, labeled diagram and calculation of CFSE, magnetic moment in case of tetrahedral complexes. (10 Marks)

(OR)

b. i. Draw the crystal field splitting diagrams in case of Ni^{2+} for an octahedral crystal field and a tetrahedral crystal field, labeling the d-orbitals. State below each diagram whether the geometry is consistent with a paramagnetic species. (6 Marks)

ii. Write a short note on spectrochemical series. (4 marks)

7 a. i Describe with suitable examples, the structural isomerism in coordination compounds. (10 Marks)

(OR)

b. i. Explain, why Cu(I) complexes such as CuI tend to be colorless, whereas Cu(II) complexes such as $Cu(NO_3)_2.5H_2O$ are brightly colored. (6 Marks)

ii. Calculate Effective Nuclear Charge (Z_{eff}) for 3d electron of Cu (Z= 29) (4 Marks)

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Set-2

INTERNAL ASSESSMENT – I (CLA1-T1)

Program: B.Tech	Date: 13-09-2023
Course Code & Title: 21CYB101J & Chemistry	Duration: 12:30 – 1:20 PM
Year &Sem: I Year & I Sem	Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 Which of the following complex compound(s) is/are paramagnetic and low spin?

(a) $K_3[Fe(CN)_6]$ (b) $[Ni(CO)_4]^0$

(c) $[Cr(NH_3)_6]^{3+}$ (d) $[Mn(CN)_6]^{4-}$

2 An aqueous solution of Ni^{2+} contains $[Ni(H_2O)_6]^{2+}$ and its magnetic moment is 2.83 BM. When ammonia is added in it the magnetic moment of solution

(a) will remain the same (b) will increase from 2.83 BM

(c) will decrease from 2.83 BM (d) cannot be predicted theoretically.

3 Which of the following is true about the complex $[PtCl_2(H_2O)(NH_3)]$?

(a) It's geometry is tetrahedral (b) It is paramagnetic complex

(c) It exhibits geometrical isomerism (d) Pt is sp^3 hybridized

4 $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$ differ in:

(a) Geometry and magnetic moment (b) Geometry and hybridization

(c) Magnetic moment and color (d) Hybridization and number of d electrons

5 The electronic configuration of metal atom/ionic octahedral complex with d^4 configuration, if $\Delta_o <$ pairing energy is:

(a) $t_{2g}^4 e_g^0$ (b) $e_g^4 t_{2g}^0$ (c) $t_{2g}^3 e_g^1$ (d) $e_g^2 t_{2g}^2$

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Part – B (2 x 10 = 20 Marks)

6. a. i. What are different types of isomerism in coordination compounds? Describe geometrical and optical isomerism with suitable examples. (10 Marks)

(OR)

b. i. Explain, why the transition metal coordination compounds with strong field ligands are yellow, orange or red in color, whereas with weak field ligands they are often blue-green, blue or indigo in color. (5 Marks)

ii. Draw structures for linkage isomers of $[Co(NH_3)_5NO_2]Cl_2$ and optical isomers of $[CoCl_2(en)_2]^+$. (5 Marks)

7 a. Calculate CFSE values in terms of Δ_o and P for high spin and low spin octahedral complexes of Fe(II) and Co(II). Predict whether the complexes are paramagnetic or diamagnetic. (10 Marks)

(OR)

b. i. Explain the factors affecting crystal field splitting with suitable examples. (6 Marks)

ii. Calculate the effective nuclear charge of a electron in 3p orbital of Aluminium atom (Atomic number of Al = 13) (4 Marks)

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SET – II

INTERNAL ASSESSMENT – I (CLA1-T1)

Program: B.Tech
Course Code & Title:21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 13/09/2023
Duration: 08:00 – 08:50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1. Among the following complexes, the one which shows zero crystal field stabilization energy (CFSE) is

   (a) $[Mn(H_2O)_6]^{3+}$ (b) $[Fe(H_2O)_6]^{3+}$ (c) $[Co(H_2O)_6]^{2+}$ (d) $[Co(H_2O)_6]^{3+}$

2. In which of the following octahedral complexes of chromium, will the magnitude of Δ_O be the lowest?

   (a) $[CrCl_6]^{3-}$ (b) $[Cr(H_2O)_6]^{3+}$ (c) $[Cr(NH_3)_6]^{3+}$ (d) $[Cr(CN)_6]^{3-}$

3. The Sc^{3+} complex is colourless in nature which is mainly due to

   (a) Partially filled d-orbitals (b) Completely filled d-orbitals

   (c) Empty d-orbitals (d) Half filled d-orbitals

4. The spin only magnetic moment of $K_3[Mn(NO_2)_6]$ is .....

   (a) ~4.9 (b) ~5.9 (c) ~1.73 (d) ~2.83

5. Shielding constant for a 2p electron in Nitrogen atom is _____

   (a) 3.1 (b) 2.1 (c) 3.45 (d) 3.9

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Part – B (2 x 10 = 20 Marks)

6. a. Based on CFT, find out the number of unpaired electrons for a Mn^{2+} complex in strong and weak octahedral field. Calculate CFSE (Δ_o) and magnetic moment for both the situation. Explain your answer with appropriate energy level diagrams. (10 Marks)

(OR)

b. i. Elucidate which of the following complexes in each pair given below will have large crystal field splitting of d-orbitals (4 Marks)

(a). $[Fe(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{3+}$ (b). $[Mn(H_2O)_6]^{2+}$ and $[Mn(CN)_6]^{4-}$

ii. Discuss in detail any two factors that influence the magnitude of crystal field splitting, Δo. (6 Marks)

7 a. i Predict the possible type of isomerism that can exist for the octahedral complex having the formula $CoBrSO_4.5NH_3$. Write down the isomers and elucidate how the isomers can be differentiated? (6 Marks)

ii. Calculate Z* for a 4s electron in Iron (Z of Iron is 26) (4 Marks)

(OR)

b. i. Draw the crystal field splitting diagrams for an octahedral crystal field and a tetrahedral crystal field of Cu^{2+} coordination complex. Label the d-orbitals and comment on the magnetic behaviour. (6 Marks)

ii. Which of the following complexes will exhibit maximum attraction to an applied magnetic field? Rationalise your answer. (4 Marks)

A. $[Ni(H_2O)_6]^{2+}$

B. $[Co(H_2O)_6]^{2+}$

C. $[Co(en)_3]^{3+}$

D. $[Zn(H_2O)_6]^{2+}$

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Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
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Date: 11/09/2023
Duration: 12.30-1.20pm
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 When the valence d orbitals of the central metal ion in octahedral complex are split in energy levels in CFT, which orbitals are raised to higher energy?

(a) d_{xy} and $d_{x^2-y^2}$ (b) d_{xy}, d_{xz} and d_{yz}

(c) d_{xz} and d_{yz} (d) $d_{x^2-y^2}$ and d_{z^2}

2 The crystal field splitting energy (Δo) is inversely proportional to

(a) geometry (b) number of d-electrons

(c) coordination number (d) oxidation state

3 In which of the following octahedral complexes of chromium, will the magnitude of Δo be the lowest?

(a) $[CrCl_6]^{3-}$ (b) $[Cr(H_2O)_6]^{3+}$

(c) $[Cr(NH_3)_6]^{3+}$ (d) $[Cr(CN)_6]^{3-}$

4 How many geometrical isomers are possible for $[Co(NH_3)_3(NO_2)_3]$ complex

a) 2 b) 3

c) 4 d) 0

5 The number of unidentate ligands in the complex ion is called

a) EAN b) Coordination number

c) Primary valency d) Oxidation number

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Part – B (2 x 10 = 20 Marks)

6. a. i. Calculate the CFSE of d^4 and d^7 in high spin tetrahedral complexes in terms of Δo (6 Marks)

ii. Write short note on linkage and hydrate isomerism in coordination compounds. (4 Marks)

(OR)

b. i. Calculate the energy of the $[Ti(H_2O)_6]^{3+}$complex with a wave number of 20,000 cm^{-1} ? (5 Marks)

ii. Which among the following complexes have large crystal field splitting in each pair with appropriate justification? (5 Marks)

(1) $[Co(H_2O)_6]^{2+}$ and $[Co(H_2O)_6]^{3+}$ (2) $[Co(CN)_6]^{3+}$ and $[Co(NH_3)_6]^{3+}$

7 a. Give the steps for Slater's rule and using it, calculate Zeff for an electron residing in 2p level for F^- ion (Z for Fluorine atom is 9).

(OR)

b. i. Draw the cis and trans isomers of $[Co(NH_3)Br(en)_2]^{2+}$ complexes. (5 Marks)

ii. Calculate the magnetic moment of $(Et_4N)_2[FeCl_4]^{2-}$ and $[Ni^{2+}(H_2O)_6]Cl_2$ complexes. (5 Marks)

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Set-2

INTERNAL ASSESSMENT – I (CLA1-T1)

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 11/09/2023
Duration: 12.30-1.20pm
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 The non-axial d-orbitals are

(a) d_{z^2} and $d_{x^2-y^2}$ (b) d_{xy} and d_{xz}

(c) $d_{x^2-y^2}$ and d_{xz} (d) d_{z^2} and d_{xz}

2 The electronic transition of octahedral $[Ti(H_2O)_6]^{3+}$ is

(a) d-d transition (b) p-p transition

(c) s-s transition (d) f-f transition

3 The CFSE for d^3 configuration of the central metal ion in tetrahedral complex is

(a) -1.2 Δo (b) -0.8 Δo

(c) -0.36 Δo (d) -1.6 Δo

4 Between these two complexes $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ which of the following statement is correct

(a) nickel is in the same oxidation state in both

(b) both have tetrahedral geometry

(c) both have square planar geometry

(d) both have tetrahedral and square planar geometry respectively

5 The magnetic moment of strong field chromium (III) octahedral complex is

(a) 1.41 BM (b) 2.83 BM

(c) 5.46 BM (d) 3.87 BM

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Part – B (2 x 10 = 20 Marks)

6. a. Discuss spectrochemical series and its importance? Explain the influence of cyanide and amine ligands on octahedral chromium (II) complexes and Δ_o variation.

(OR)

b. i. Write short note on structural and ionization isomerism in coordination compounds. (5 Marks)

ii. Calculate the shielding constant and effective nuclear charge for a valence electron in fluorine. (5 Marks)

7 a. i Calculate the wavelength of the light absorbed by $[Ni(H_2O)_6]^{2+}$ complex with a wave number of 25,000 cm^{-1}? (5 Marks)

ii. Calculate the magnetic moment of $K_3[Fe(oxalate)_3]$ $3H_2O$ and K_2CuCl_4 complexes. (5 Marks)

(OR)

b. i. Draw the cis and trans isomers of $[Co(en)_2Cl_2]$ complexes. (5 Marks)

ii. Explain briefly about high spin and low spin complexes with examples. (5 Marks)

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SET – 1

INTERNAL ASSESSMENT – I (CLA1-T1)

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: 1 Year & I Sem

Date: 11-09-2023
Time: 8.00-8.50am
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. The crystal field splitting energy of an octahedral complex is 18,000 cm^{-1}. What will be the energy in terms of Δ_t ?
   a) 40,500 cm^{-1} b) 8100 cm^{-1} c) 36,000 cm^{-1} d) 9000 cm^{-1}

2. The number of unpaired electrons present in $[Fe(CN)_6]^{4-}$ is
   a) 0 b) 1 c) 2 d) 3

3. Arrange the following ligands, en, NH_3, H_2O and CN^- in their decreasing order of field strength
   a) $NH_3 > CN^- > H_2O >$ en
   b) en $> CN^- > H_2O > NH_3$
   c) $CN^- >$ en $> NH_3 > H_2O$
   d) $CN^- >$ en $> H_2O > NH_3$

4. The complexes $[Co(NH_3)_6][Cr(C_2O_4)_3]$ and $[Cr(NH_3)_6][Co(C_2O_4)_3]$
   (a) Linkage isomerism
   (b) Geometrical isomerism
   (c) Coordination isomerism
   (d) Ionisation isomerism

5. The shielding constant of 1s electron in helium atom is_____
   a) 1.70 b) 0.35 c) 1.65 d) 0.30

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Part – B (2 x 10 = 20 Marks)

6. a. (i) Discuss the salient features of crystal field theory. (4 Marks)

(ii) Identify the complex possessing the larger value of Δ_o in the following pairs and give reason. (6 Marks)

(A) $[Ni(CN)_6]^{4-}$ and $[Ni(H_2O)_6]^{2+}$

(B) $[Co(H_2O)_6]^{2+}$ and $[Co(H_2O)_6]^{3+}$

(C) $[Rh(NH_3)_6]^{3+}$ and $[Ir(NH_3)_6]^{3+}$

(OR)

b. Discuss the crystal field splitting of a tetrahedral complex. Calculate the CFSE and magnetic moment of $[NiCl_4]^{2-}$ (Atomic number of Ni is 28). (10 Marks)

7. a. Brief about the following isomerism exhibited by the coordination compounds with a suitable example for each. (3+3+4 Marks)

(i) Ionization

(ii) Hydrate

(iii) Geometrical

(OR)

b. (i) The UV-Visible spectrum of an octahedral complex showed λmax at 600 nm. Calculate the crystal field splitting energy, Δ_o and complementary colour of the complex? (6 Marks)

(ii) Determine the effective nuclear charge experienced by the 4p electron of Bromine. [Atomic number of Br = 35]. (4 Marks)

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SET – 2

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 16-02-23
Duration: 12.30-1.20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. A magnetic moment of 1.73BM will be shown by one among the following
   a) $TiCl_4$ b) $[Ni(Cl_4)]^{2-}$ c) $[Cu(NH_3)_4]^{2+}$ d) $[Co(Cl_6)]^{4-}$

2. The ligand that produces the strongest field and causes maximum splitting is _____
   a) F^- b) H_2O c) NH_3 d) CO

3. Which type of isomerism is exhibited by $Pt(NH_3)_2Cl_2$?
   a) Coordination isomerism b) Linkage isomerism
   c) Optical isomerism d) Geometrical isomerism

4. Which of the following species has the highest ionization potential?
   a) Li^+ b) Mg^+ c) He d) Ne

5. According to HSAB concept, which of the following will prefer to exist as carbonates
   b) Mg^{2+} b) Hg^{2+} c) Pt^{2+} d) Al^{3+}

Part – B (2 x 10 = 20 Marks)

6. a. Discuss the crystal field splitting of a tetrahedral complex. Calculate the CFSE and magnetic moment of $[Mn(Cl_4)]^{2-}$ (Atomic number of Mn is 25). (10 Marks)

(OR)

b. i. Describe the salient features of Crystal Field Theory. (4 Marks)

ii. An octahedral complex absorbs light with a wavenumber of 18,700 cm^{-1}. Determine the crystal field splitting energy, Δ_o and complementary colour of the complex? (6 Marks)

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7 a. i Discuss the optical isomerism exhibited by the transition metal compounds with any two examples. (6 Marks)

ii. What is the effective nuclear charge experienced by a 3d electron in Cu [Z=29]? (4 Marks)

(OR)

b. i. Define Electronegativity? Give its variation along the period and down the group with examples. (6 Marks)

ii. Give a brief account on hard and soft acids. (4 Marks)

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Set-1

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
PM
Year & Sem: I Year & II Sem

Date: 16.02.23
Duration: 12.30 – 1.20

Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. Among the following complexes, the one which shows zero crystal field stabilization energy (CFSE) is
   a) $[Fe(H_2O)_6]^{3+}$ b) $[Mn(H_2O)_6]^{3+}$ c) $[Co(H_2O)_6]^{3+}$ d) $[Co(H_2O)_6]^{2+}$

2. How many unpaired electrons are there in a strong field iron(II) octahedral complex?
   a) 0 b) 1 c) 2 d) 4

3. Which one of the following complexes is NOT expected to exhibit isomerism?
   a) $[Ni(NH_3)_4(H_2O)_2]^{2+}$ b) $[Pt(NH_3)_2Cl_2]$
   c) $[Co(NH_3)_5SO_4]Cl$ d) $[Fe(en)_3]^{3+}$

4. Choose the correct statement
   a) As shielding effect increases electronegativity decreases
   b) As shielding effect increases electronegativity increases
   c) As shielding effect increases ionization potential increases
   d) As positive charge on species increases ionic radii increases

5. Which of the following transition metal ions will prefer to form thiocyanato (SCN^-) complex on the basis of HSAB principle?
   a) Co^{3+} b) Cr^{3+} c) Fe^{3+} d) Pt^{4+}

Part – B (2 x 10 = 20 Marks)

6. a. i. Compare the CFSE and magnetic moment values of $[Ni(CN)_6]^{4-}$ and $[Ni(H_2O)_6]^{2+}$ complexes. [Atomic number of Ni = 28] (6 Marks)

   ii. Explain why the tetrahedral complexes are always high spin? (4 Marks)

(OR)

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b. Explain in detail the d-orbital splitting in octahedral complexes and d-d transition with a suitable example. (10 Marks)

7. a. i. Brief about the geometrical isomerism exhibited by octahedral complexes. (6 Marks)

ii. Calculate the effective nuclear charge experienced by the valence electron of Zn [Z=30]. (4 Marks)

(OR)

b. i. Define the term Electron affinity. Discuss its variation across the period and down the group with examples. (6 Marks)

ii. Aluminium occurs in nature as oxide whereas Platinum occurs as sulphide. Explain based on HSAB principle. (4 Marks)

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Set-2

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 16.03.23
Duration: 8.00-8.50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1 Crystal field stabilization energy for high spin d^4 octahedral complex is:

(a) $-1.8\ \Delta_o$ (b) $-1.6\ \Delta_o + P$ c.) $-1.2\ \Delta_o$ d.) $-0.6\ \Delta_o$

2 The largest paramagnetism is shown by _____

(a) $[Cr(H_2O)_6]^{3+}$ (b) $[Fe(H_2O)_6]^{2+}$
(c) $[Cu(H_2O)_6]^{2+}$ (d) $[Zn(H_2O)_6]^{2+}$

3 What kind of isomerism(s) is(are) possible for a complex $[Co(NH_3)_4\ Br_2]Cl$?

a) geometrical and ionization
b) geometrical and optical
c) geometrical only
d) ionization only

4 Choose the correct order of ionization energy:

a) N > O > F b) N > O < F c) F > O > N d) O > F > N

5 Which of the following will prefer to exist as sulphide?

a) Hg^{2+} b) Al^{3+} c) Mg^{2+} d) Ca^{2+}

Part – B (2 x 10 = 20 Marks)

6. a. Write a short note on spectrochemical series? Explain with a suitable example, the influence of ligands in the formation of high and low spin complexes. (10 Marks)

(OR)

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b. i. Arrange the following complexes in their increasing order of wavelength of light absorbed and explain. (6 Marks)
$[Co(NH_3)_6]^{3+}$, $[Co(CN)_6]^{3-}$ and $[Co(H_2O)_6]^{3+}$

ii. Calculate the crystal field splitting energy, Δ_o for $[Cu(H_2O)_6]^{2+}$ complex at the maximum absorbance (λ_{max}) of 600nm and predict its colour. (4 Marks)

7. a. i. Discuss the hydration and linkage isomerism shown by the octahedral complexes with an example. (6 Marks)

ii. Determine the screening constant and effective nuclear charge of 2p electron in Mg^{2+} ion. [Number of protons = 12]. (4 Marks)

(OR)

b. i. Define atomic and ionic radii. Arrange the following O, O^- and O^{2-} in their increasing order of radii. (6 Marks)

ii. According to HSAB principle, predict whether the following reaction is feasible or not.

$$BeI_2 + HgF_2 \rightarrow BeF_2 + HgI_2$$ (4 Marks)

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SET – 1

INTERNAL ASSESSMENT – I

Program: B.Tech	Date: 16-02-23
Course Code & Title:21CYB101J & Chemistry	Duration: 08:00 – 08:50 AM
Year & Sem: I Year & II Sem	Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. The crystal field splitting energy for octahedral and tetrahedral complexes is related as
   a) $\Delta_t = 4/9\ \Delta_o$ b) $\Delta_t = 1/2\ \Delta_o$ c) $\Delta_o = 2\ \Delta_t$ d) $\Delta_o = 4/9\ \Delta_t$

2. Which of the following octahedral complexes of Co (atomic no.27) will have the highest Δ_o value?
   a) $[Co(CN)_6]^{3-}$ b) $[Co(C_2O_4)_3]^{3-}$ c) $[Co(H_2O)_6]^{3+}$ d) $[Co(NH_3)_6]^{3+}$

3. Which of the following species will be diamagnetic?
   a) $[Fe(CN)_6]^{4-}$ b) $[FeF_6]^{3+}$ c) $[Co(C_2O_4)_3]^{3-}$ d) $[CoF_6]^{3-}$

4. The screening constant of 2s electron in lithium atom is_____
   a) 0.85 b) 0.35 c) 1.20 d) 1.70

5. According to HSAB concept, which of the following will prefer to occur as oxides
   a) Ag^+ b) Cu^{2+} c) Pt^{2+} d) Al^{3+}

Part – B (2 x 10 = 20 Marks)

6. a. i Which of the following complexes have large crystal field splitting of d-orbitals in each pair and why? (4 Marks)

   (A). $[Co(H_2O)_6]^{2+}$ and $[Ni(H_2O)_6]^{2+}$ (B) $[V(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{3+}$

   ii. Calculate the CFSE and magnetic moment values for the high and low spin octahedral complexes of Co(II) [Z = 27] in terms of Δ_o and P. Predict whether the complexes are paramagnetic and diamagnetic. (6 Marks)

   (OR)

   b. i. Discuss any two factors that influence the magnitude of crystal field splitting, Δ_o. (6 Marks)

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ii. $[Ti(H_2O)_6]^{3+}$ is coloured, while $[Sc(H_2O)_6]^{3+}$ is colourless – Explain. [Note: Atomic number of Sc and Ti is 21 and 22 respectively]. (4 Marks)

7 a. i Elaborate the ionization and coordination isomerism shown by the octahedral complexes with an example. (6 Marks)

ii. Calculate the effective nuclear charge of a 3p electron in the chlorine atom. (Number of Protons = 17). (4 Marks)

(OR)

b. i. Define ionization potential? Give its variation along the period and down the group with examples. (6 Marks)

ii. Write a note on the application of HSAB principle in explaining the stability of complexes with the examples. (4 Marks)

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Set-1

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 13-02-23
Duration: 12:30 – 1:20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1 $[Cr(CN)_6]^{3-}$ is expected to be: (Z of Cr = 24)

(a) paramagnetic with $\mu_{eff} \approx 3.87$ BM (b) diamagnetic

(c) paramagnetic with $\mu_{eff} < 3.87$ BM (d) paramagnetic with $\mu_{eff} > 3.87$ BM

2 For which pair of complexes is the order of values of Δ_{oct} correct?

(a) $[Rh(NH_3)_6]^{3+} > [Co(NH_3)_6]^{3+}$

(b) $[Fe(CN)_6]^{4-} > [Fe(CN)_6]^{3-}$

(c) $[Cr(OH_2)_6]^{2+} > [Cr(OH_2)_6]^{3+}$

(d) $[CrF_6]^{3-} > [Cr(CN)_6]^{3-}$

3 What is the secondary valency of metal ion in $[Cr(NH_3)_4(en)]NO_3$?

(a) 2 (b) 4 (c) 6 (d) 8

4 Which of the following is the correct order of elements with increasing electronegativity?

(a) S < P < N < O

(b) N < O < P < S

(c) N < O < P < S

(d) P < S < N < O

5 Which of the following metals are usually found in the form of oxide ores?

(a) Ag^+ and Hg^{2+} (b) Fe^{3+} and Al^{3+}

(c) Ag^+ and Fe^{3+} (d) Hg^{2+} and Al^{3+}

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Part – B (2 x 10 = 20 Marks)

6. a. Explain crystal field splitting of d orbitals in octahedral geometry and also give the calculation of CFSE and magnetic moment for octahedral complexes? (10 Marks)

(OR)

b. i. For $[Cr(H_20)_6]^{2+}$ ion, the mean pairing energy (P) is found to be equal to 23500 cm^{-1}. The magnitude of Δ_o is 13900 cm^{-1}. Calculate the CFSE for this complex ion corresponding to high spin and low spin state. (6 Marks)

ii. Draw the structures for linkage isomers of $[FeCl_5SCN]^{3-}$ and optical isomers of $[PtCl_2(en)_2]^{2+}$ (4 Marks)

7 a. i Explain HSAB principle. Give its applications and limitations. (10 Marks)

(OR)

b. i. The octahedral complex $[Ti(H_2O)_6]^{3+}$ has a single d electron. To excite this electron from the ground state t_{2g} orbital to the eg orbital, this complex absorbs light from 450 to 600 nm. The maximum absorbance corresponds to Δ_o and occurs at 499 nm. Calculate the value of Δ_o in Joules and predict the perceived color of the solution. ($h = 6.626 \times 10^{-34}$ Js, $c = 2.99 \times 10^8$ m/s) (6 Marks)

ii. In a second period, the ionization energy is expected to vary in following order Li < Be < B < C < N < O < F < Ne, however the observed trend is Li < B < Be < C < O < N < F < Ne. Give reasons for its discrepancy. (4 Marks)

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Set-2

INTERNAL ASSESSMENT – 1

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 13-02-23
Duration: 12:30 – 1:20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1 The co-ordination number, oxidation number, number of d electrons and the number of unpaired electrons for the central metal atom in the complex $[Co(en)_2(NH_3)_2]^{2+}$ are ----- respectively

(a) 4,2,7,1 (b) 6,2,6,1
(c) 6,2,7,1 (d) 4,2,6,3

2 What is theoretical magnetic moment of complex compound $K_2[NiF_4]$?

(a) 1.73 BM (b) 2.83 BM (c) 3.87 BM (d) zero

3 Among SCN^-, NO_3^-, NO_2^- and $C_2O_4^{2-}$, which are ambidentate ligands?

(a) SCN^- and NO_2^- (b) NO_3^- and NO_2^-
(c) NO_2^- and $C_2O_4^{2-}$ (d) SCN^- and $C_2O_4^{2-}$

4 Which is smallest in size?

(a) O^{2-} (b) C^{4-} (c) F^- (d) N^{3-}

5 According to hard and soft acid base principle, a hard acid:

a) has low charge density
b) shows preference for soft bases
c) shows preference for donor atoms of low electronegativity
d) is not polarizable

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Part – B (2 x 10 = 20 Marks)

6. a. i. Describe stereo-isomerism in co-ordination compounds with suitable examples. (10 Marks)

(OR)

b. i. Ni^{2+} in a coordination complex is paramagnetic. Draw the crystal field splitting diagrams for an octahedral crystal field and a tetrahedral crystal field, labeling the d-orbitals. State below each diagram whether the geometry is consistent with a paramagnetic species. (6 Marks)

ii. The strong ligand complexes are usually yellow, orange or red in color while weak ligand field complexes are blue, green or indigo in color, explain. (4 Marks)

7 a. Define and explain the following periodic properties :

(A) Van der Waals radius (B) Electron affinity

(C) Ionization energy (D) Electronegativity (10 Marks)

(OR)

b. i. Calculate Effective Nuclear Charge (Z_{eff}) for 3d electron of Cr (Z= 24) (6 Marks)

ii. Predict and explain the products of the following reaction:
LiI + CsF ---------- ? (4 Marks)

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College of Engineering and Technology
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INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 13-02-23
Duration: 8.00 – 8.50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. Which of the following complex is paramagnetic? (Z of Cu = 29, Fe = 26, Co = 27)

   (a) $[Cu(CN)_4]^{3-}$ (b) $[Fe(CN)_6]^{4-}$ (c) $[CoBr]_4^2$ (d) $[Co(NH_3)_6]^{3+}$

2. The correct increasing order of splitting power of ligands according to spectrochemical series is

   (a) $Cl^- < OH^- < CN^-$ (b) $Cl^- < CN^- < OH^-$
   (c) $OH^- < Cl^- < CN^-$ (d) $OH^- < CN^- < Cl^-$

3. Which one of the following ions exhibits colour in aqueous solution?

   (a) Sc^{3+} (b) Ni^{2+} (c) Ti^{4+} (d) Zn^{2+}

4. Which of the following order is correct for the first ionization energies of following elements?

   (a) B < Be < N < O (b) Be < B < N < O
   (c) B < Be < O < N (d) B < O < Be < N

5. According to HSAB theory, which option shows negative correlation with hardness?

   (a) Oxidation State (b) Polarizability
   (c) Electronegativity (d) Charge density

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Part – B (2 x 10 = 20 Marks)

6. a. Explain crystal field splitting of d orbitals, the calculation of CFSE and magnetic moment in the case of tetrahedral complexes? (10 Marks)

(OR)

b. i. For Mn^{2+} ion, P = 28,000 cm^{-1} and Δ_o value for $[Mn(CN)_6]^{3-}$ ion is 38,500 cm^{-1}. Predict whether the complex is high spin or low spin. Write the configuration and calculate CFSE in cm^{-1}. (6 Marks)

ii. Write the formula for ionization isomer of $[Co(en)_2Cl.SCN]NO_2$ and hydration isomer of $[Cr(H_2O)_6]Cl_3$ (4 Marks)

7 a. Explain HSAB principle. Give its applications and limitations. (10 Marks)

(OR)

b. i. Calculate the octahedral crystal field splitting energy in kJ for $[Fe(CN)_6]^{4-}$, if the wavelength of the most intensely absorbed light is 305 nm.
($h = 6.626 \times 10^{-34}$ Js, $c = 2.99 \times 10^8$ m/s) (6 Marks)

ii. Atomic size decreases from Li to F in a second period; however from F to Ne it increases. Give reasons. (4 Marks)

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SET – 2

INTERNAL ASSESSMENT – I

Program: B.Tech	**Date: 13-02-23**
Course Code & Title: 21CYB101J & Chemistry	**Duration: 8.00 – 8.50 AM**
Year & Sem: I Year & II Sem	**Max. Marks: 25 marks**

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. Which of the following complexes has a magnetic moment of 1.73 BM?
(Z of Ni = 28, Ti = 22, Cu = 29, Co = 27)

(a) $[Ni(CN)_4]^{2-}$ (b) $TiCl_4$ (c) $[Cu(NH_3)_4]^{2+}$ (d) $[CoCl_6]^{4-}$

2. The correct electronic configuration of the central atom in $K_4[Fe(CN)_6]$ based on crystal field theory is

(a) $e^4\ t_2^2$ (b) $t_{2g}^4\ e_g^2$ (c) $t_{2g}^6\ e_g^0$ (d) $e_3\ t_2^3$

3. Out of TiF_6^{2-}, CoF_6^{3-}, $Fe(CN)_6^{4-}$ and $NiCl_4^{2-}$ (Z of Ti = 22, Co = 27, Fe = 26, Ni = 28), the colourless species is :

(a) TiF_6^{2-} (b) CoF_6^{3-} (c) $Fe(CN)_6^{4-}$ (d) $NiCl_4^{2-}$

4. The electron affinity for the inert gases is

(a) 0 (b) High (c) Negative (d) Positive

5. According to hard and soft acid base principle, hard acid

(a) is not polarizable
(b) has low charge density
(c) shows preference for donor atoms of low electronegativity
(d) shows preference for soft bases

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Part – B (2 x 10 = 20 Marks)

6. a. Describe structural isomerism in co-ordination compounds with suitable examples. (10 Marks)

(OR)

b. i. Calculate CFSE values in terms of Δ_o and P for high and low spin octahedral complexes of Fe (II). Predict whether the complexes are paramagnetic or diamagnetic? (4 Marks)

ii. Which of the following complexes have large crystal field splitting of d-orbitals in each pair and why?

(a) $[Fe(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{3+}$ (b) $[Co(H_2O)_6]^{2+}$ and $[Ni(H_2O)_6]^{2+}$

(c) $[Co(NH_3)_6]^{3+}$ and $[Rh(NH_3)_6]^{3+}$ (6 Marks)

7 a. i Define 'Ionization Energy'. Discuss the factors affecting it and its variation across the period and down the group in a periodic table. (10 Marks)

(OR)

b. i. Calculate Effective Nuclear Charge (Z_{eff}) for 3d electron of V (Z= 23) (6 Marks)

ii. Explain complex formation of SCN^- with Pt^{2+} and Cr^{3+}. (4 Marks)

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SET – II

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 11/10/2022
Duration: 08:00 – 08:50 AM
Max. Marks: 25 marks

Part -- A (5 x 1 = 5 Marks)

Answer ALL The Questions

1. The crystal field splitting energy (Δo) is directly proportional to
   (a) geometry (b) number of d-electrons
   (c) coordination number (d) oxidation state
2. In spectrochemical series, the order of ligands is
   (a) π acceptor < π donor < σ donor (b) σ acceptor < π donor < σ donor
   (c) π donor < σ donor < π acceptor (d) π acceptor < σ donor < π donor
3. The Sc^{3+} complex is colourless in nature which is mainly due to
   (a) Partially filled d-orbitals (b) Completely filled d-orbitals
   (c) Empty d-orbitals (d) Half filled d-orbitals
4. The second ionisation energy is always higher than the first ionization energy because the
   (a) electron is attracted more by the core electrons
   (b) electron is more tightly bound to the nucleus in an ion
   (c) becomes more stable attaining the octet or duplet configuration
   (d) atomic radii is large
5. All cations and molecules that are short of an electron pair act as _____
   (a) Lewis acids (b) Lewis bases (c) Bronsted acid (d) Bronsted bases

Part – B (2 x 10 = 20 Marks)

6. a. Find out the number of unpaired electrons in strong and weak octahedral field for Mn^{2+} complex based on CFT. Calculate CFSE and magnetic moment for both the situation with energy level diagrams. (10 Marks)

(OR)

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b. i. Which of the following complexes have large crystal field splitting of d-orbitals in each pair and why? (5 Marks)

(a). $[Fe(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{3+}$ (b). $[Fe(CN)_6]^{3-}$ and $[Fe(NH_3)_6]^{3+}$

ii. Calculate the wavelength of the light absorbed by octahedral complex with a frequency of 24,000 cm^{-1}? (5 Marks)

7 a. i Predict the possible type of isomerism can exist for the octahedral complex when the central metal ion is Co^{2+} surrounded by 5 NH_3 molecule, 1 Br^- and 1 SO_4^{2-} ions. Write down the isomers and how to test the difference among the isomers? (6 Marks)

ii. Explain why electron affinity of chlorine is more than that of fluorine. (4 Marks)

(OR)

b. i. Why Cl^- ion will be bigger than Cl (Atomic Number = 17)? Justify using Slater's Rule. (6 Marks)

ii. Give the differences between hard and soft acids. (4 Marks)

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SET – I

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 11/10/2022
Duration: 08:00 – 08:50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1. CFSE for $t_{2g}^p e_g^q$ configuration of octahedral complex is
   (a) $[-0.4p + 0.6q]\ \Delta_o$ (b) $[-0.27p + 0.18q]\ \Delta_o$
   (c) $[-0.18p + 0.27q]\ \Delta_o$ (d) $[-0.5p + 0.5q]\ \Delta_o$
2. Among the following, which is the π donor ligand?
   (a) H_2O (b) en (c) CO (d) Cl^-
3. In which of the following octahedral complexes of chromium, will the magnitude of Δ_O be the lowest?
   (a) $[CrCl_6]^{3-}$ (b) $[Cr(H_2O)_6]^{3+}$ (c) $[Cr(NH_3)_6]^{3+}$ (d) $[Cr(CN)_6]^{3-}$
4. The effective nuclear charge realised by 1s electron of helium atom is___
   (a) 0.70 (b) 0.30 (c) 2.00 (d) 1.70
5. Which of the following metal ions form sulphides?
   (a) Ca^{2+} and Al^{3+} (b) Ag^+ and Hg^{2+} (c) Ca^{2+} and Ag^+ (d) Al^{3+} and Hg^{2+}

Part – B (2 x 10 = 20 Marks)

6. a. The energy gap between t_{2g} and e_g of an octahedral complex is 239.23 kJ/mol (1kJ/mol = 83.6 cm^{-1}). Find the wavelength absorbed by the complex for d-d transition and predict colour of the sample based on complementary colour. (10 Marks)

(OR)

b. i. Write down the salient features of Crystal Field Theory. (4 Marks)

ii. Determine CFSE and magnetic moment of a d^6 octahedral complex having Δ_O = 25000 cm^{-1} and P= 15000 cm^{-1}. (6 Marks)

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7 a. i Calculate the magnetic moment of $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ complexes. (4 Marks)

ii. What is the shielding constant experienced by a 3d electron in the bromine atom (Number of Protons = 35)? Calculate the effective nuclear charge for the same. (6 Marks)

(OR)

b. i. Define atomic radii. Give its variation along the period and down the group taking examples. Arrange the following in the increasing order of atomic radii and give reasons: N, S, P and O. (5 Marks)

ii. Explain complex formation of thiocyanate ligand with Fe^{3+} and Au^{+} ions using HSAB concept. (5 Marks)

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SET – II

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 11/10/2022
Duration: 12:30 – 01:20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1. Among the following complexes, the one which shows zero crystal field stabilization energy (CFSE) is

   (a) $[Mn(H_2O)_6]^{3+}$ (b) $[Fe(H_2O)_6]^{3+}$ (c) $[Co(H_2O)_6]^{2+}$ (d) $[Co(H_2O)_6]^{3+}$

2. The crystal field splitting energy (Δo) is inversely proportional to

   (a) geometry (b) number of d-electrons

   (c) coordination number (d) oxidation state

3. The electronic transition of octahedral $Ti(H_2O)_6]^{3+}$ is

   (a) d-d transition (b) p-p transition (c) s-s transition (d) f-f transition

4. Choose the correct statement

   (a) As shielding effect increases electro negativity decreases

   (b) As shielding effect increases electro negativity increases

   (c) As +ve charge on species increases ionic radii increases

   (d) As ionization potential increases metallic property increases

5. Identify the hard acid from the following:

   (a)$AlCl_3$ (b)N_2H_4 (c)H_2O (d)OH^-

Part – B (2 x 10 = 20 Marks)

6. a. The energy gap between t_{2g} and e_g of an octahedral complex is 259 kJ/mol (1kJ/mol = 83.6 cm^{-1}). Find the wavelength absorbed by the complex for d-d transition and predict colour of the sample based on complementary colour. (10 Marks)

   (OR)

   b. i. Write down the limitations of Crystal Field Theory. (4 Marks)

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ii. Which of the following complexes have large crystal field splitting of d-orbitals in each pair and why? (6 Marks)

(a). $[Mn(H_2O)_6]^{2+}$ and $[Mn(H_2O)_6]^{3+}$ (b). $[Mn(CN)_6]^{3-}$ and $[Mn(NH_3)_6]^{3+}$

7 a. i Calculate the magnetic moment of $[Fe(NH_3)_6]^{3+}$ and $[FeF_6]^{3-}$ complexes. (4 Marks)

ii. What is the shielding constant experienced by 6s electron in the gold atom (Number of Protons = 79)? Calculate the effective nuclear charge for the same. (6 Marks)

(OR)

b. i. What do you understand by the term electronegativity? List out the elements from the following the most electropositive and electronegative element.

Li, Be, B, C, K and F. (6 Marks)

ii. Give the differences between hard and soft base. (4 Marks)

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Set-1

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 14/10/2022
Duration: 08:00 – 08:50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 How many unpaired electrons are present in a weak field iron (II) octahedral complex in CFT?

(a) 5 (b) 1

(c) 2 (d) 4

2 The axial d-orbitals are

(a) d_z^2 and $d_{x^2-y^2}$ (b) d_{xy} and d_{xz}

(c) d_{yz} and d_{xz} (d) d_z^2 and d_{xz}

3 The crystal field splitting energy (Δo) is directly proportional to

(a) geometry (b) number of d-electrons

(c) shape of the d-orbitals (d) oxidation state

4 Identify the increasing order of spectrochemical series

(a) $I^- < Br^- < S^{2-} < en < NO^{2-} < CN^- < CO$ (b) $I^- < Br^- < S^{2-} < en < NO^{2-} < CO < CN^-$

(c) $CO > CN^- > NO^{2-} > en > I^- < Br^- < S^{2-}$ (d) $I^- < Br^- < S^{2-} = en = NO^{2-} = CN^- < CO$

5 Identify the reagent, among the following, which one will provide distinction between the ionisation isomers of the formula $[Co(NH_3)_5Br]SO_4$.

(a) $BaCl_2$ (b) HCl

(c) H_2SO_4 (d) Fenton's reagent

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Part – B (2 x 10 = 20 Marks)

6. a. Compare the crystal field splitting in octahedral and tetrahedral complexes with a neat sketch by taking an example.

(OR)

b. i. Which among the following complexes have large crystal field splitting in each pair with appropriate justification? (6 Marks)

(1) $[Co(H_2O)_6]^{2+}$ and $[Co(H_2O)_6]^{3+}$ (2) $[Co(CN)_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$

ii. Write short note on structural and ionization isomerism in coordination compounds. (4 Marks)

7 a. i Calculate the shielding constant and effective nuclear charge for a valence electron in potassium (Z=19). (6 Marks)

ii. Calculate the wavelength of the light absorbed by $[Ni(H_2O)_6]^{2+}$ complex with a frequency of 25,000 cm^{-1}. (4 Marks)

(OR)

b. i. Explain briefly about high spin and low spin complexes with examples. (5 Marks)

ii. Draw the cis and trans isomers of $[Co(NH_3)Br(en)_2]^{2+}$ complexes. (5 Marks)

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Set-1

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 14/10/2022
Duration: 12:30 – 1:20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 The non-axial d-orbitals are

(a) d_{z^2} and $d_{x^2-y^2}$ (b) d_{xy} and d_{xz}

(c) $d_{x^2-y^2}$ and d_{xz} (d) d_{z^2} and d_{xz}

2 The electronic transition of octahedral $[Ti(H_2O)_6]^{3+}$ is

(a) d-d transition (b) p-p transition

(c) s-s transition (d) f-f transition

3 The CFSE for d^3 configuration of the central metal ion in tetrahedral complex is

(a) -1.2 Δo (b) -0.8 Δo

(c) -0.36 Δo (d) -1.6 Δo

4 Between these two complexes $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ which of the following statement is correct

(a) nickel is in the same oxidation state in both

(b) both have tetrahedral geometry

(c) both have square planar geometry

(d) both have tetrahedral and square planar geometry respectively

5 Among the following which is least soluble in water

(a) $CaSO_4$ (b) $MgSO_4$

(c) Na_2SO_4 (d) $BaSO_4$

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Part – B (2 x 10 = 20 Marks)

6. a. Discuss spectrochemical series and its importance? Explain the influence of cyanide and amine ligands on octahedral chromium (II) complexes and Δ_o variation.

(OR)

b. i. Explain with examples the variation of any two periodic properties along the period and down the group. (5 Marks)

ii. Calculate the shielding constant and effective nuclear charge for a valence electron in potassium (Z=19). (5 Marks)

7 a. i Calculate the wavelength of the light absorbed by $[Ni(H_2O)_6]^{2+}$ complex with a frequency of 25,000 cm^{-1}? (5 Marks)

ii. Calculate the magnetic moment of $K_3[Fe(oxalate)_3]$ $3H_2O$ and K_2CuCl_4 complexes. (5 Marks)

(OR)

b. i. Draw the cis and trans isomers of $[Co(en)_2Cl_2]$ complexes. (4 Marks)

ii. Give the differences between hard and soft acids with examples. (6 Marks)

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Set-2

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 14/10/2022
Duration: 12:30 – 1:20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 When the valence d orbitals of the central metal ion in octahedral complex are split in energy levels in CFT, which orbitals are raised to higher energy?

(a) d_{xy} and $d_{x^2-y^2}$ (b) d_{xy}, d_{xz} and d_{yz}

(c) d_{xz} and d_{yz} (d) $d_{x^2-y^2}$ and d_{z^2}

2 The crystal field splitting energy (Δo) is inversely proportional to

(a) geometry (b) number of d-electrons

(c) coordination number (d) oxidation state

3 In which of the following octahedral complexes of chromium, will the magnitude of Δ_o be the lowest?

(a) $[CrCl_6]^{3-}$ (b) $[Cr(H_2O)_6]^{3+}$

(c) $[Cr(NH_3)_6]^{3+}$ (d) $[Cr(CN)_6]^{3-}$

4 How many geometrical isomers are possible for $[Co(NH_3)_3(NO_2)_3]$ complex

a) 2 b) 3

c) 4 d) 0

5 The number of unidentate ligands in the complex ion is called

a) EAN b) Coordination number

c) Primary valency d) Oxidation number

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Part – B (2 x 10 = 20 Marks)

6. a. i. Calculate the CFSE of d^4 and d^7 in high spin tetrahedral complexes in terms of Δo (6 Marks)

ii. Write short note on linkage and hydrate isomerism in coordination compounds. (4 Marks)

(OR)

b. i. Calculate the energy of the $[Ti(H_2O)_6]^{3+}$complex with a frequency of 20,000 cm^{-1} ? (6 Marks)

ii. Give the differences between hard and soft bases. (4 Marks)

7 a. Using Slater's rule explain the filling of 4s - orbital prior to the filling of 3d - orbital in the case of Potassium (Z = 19)?

(OR)

b. i. Explain why the first ionization energy of Al is lower than Mg? (5 Marks)

ii. Calculate the magnetic moment of $(Et_4N)_2[FeCl_4]^{2-}$ and $[Ni^{2+}(H_2O)_6]Cl_2$ complexes. (5 Marks)

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SRM

DEPARTMENT OF CHEMISTRY
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

SET – I

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 11/10/2022
Duration: 12:30 – 01:20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1. In CFT, the weak field ligand such as Cl^- usually produces:
   (a) high spin complexes and small crystal field splitting
   (b) low spin complexes and small crystal field splitting
   (c) low spin complexes and high crystal field splitting
   (d) high spin complexes and high crystal field splitting
2. CFSE for $e^p t_2^q$ configuration of tetrahedral complex is
   (a) $[-0.4p + 0.6q]\ \Delta_o$ (b) $[-0.27p + 0.18q]\ \Delta_o$
   (c) $[-0.6p + 0.4q]\ \Delta_o$ (d) $[-0.5p + 0.5q]\ \Delta_o$
3. How many unpaired electrons are present in a strong field Mn(II) octahedral complex?
   (a) 0 (b) 1 (c) 2 (d) 4
4. Choose the correct order ionization energy
   (a) N>O>F (b) F>O>N (c) N>O<F (d) O>F>N
5. For hard acids, K_f= F- >> Cl- > Br- > I-
   (a) False (b) reverse is true (c) For soft acids (d) True

Part – B (2 x 10 = 20 Marks)

6. a. Find out the number of unpaired electrons in strong and weak octahedral field for Fe^{3+} complex based on CFT. Calculate CFSE and magnetic moment for both the situation with energy level diagrams. (10 Marks)

(OR)

b. i. Determine CFSE and magnetic moment of a d^7 octahedral complex having Δ_O = 13000 cm^{-1} and P= 21000 cm^{-1}. (6 Marks)

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ii. Calculate the wavelength of the light absorbed by octahedral complex with a frequency of 17,830 cm^{-1} ? (4 Marks)

7. a. i. Predict the possible type of isomerism can exist for the octahedral complex $CrCl_3.6H_2O$. Write down the isomers and how to test the difference between the isomers ? (6 Marks)

ii. Why electron affinity decreases from C to N in the second period? (4 Marks)

(OR)

b. i. Why Li ion will be bigger than Li^+ (Atomic Number = 3)? Justify using Slater's Rule? (6 Marks)

ii. Explain feasibility of the reaction between the following compounds using HSAB concept. (4 Marks)

(a). LiI and CsF (b) CaS and H_2O

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SRM

DEPARTMENT OF CHEMISTRY

SET – 2

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

INTERNAL ASSESSMENT – I (CLAI-T1)

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: 1 Year & 1 Sem

Date: 11-09-2023
Duration: 8.00-8.50am
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. Among the following complexes, the one which shows zero crystal field stabilization energy (CFSE) is
   a) $[Fe(H_2O)_6]^{3+}$ b) $[Mn(H_2O)_6]^{3+}$ c) $[Co(H_2O)_6]^{3+}$ d) $[Co(H_2O)_6]^{2-}$

2. The complex possessing high magnetic moment value is _____
   (a) $[Cr(H_2O)_6]^{3+}$ (b) $[Fe(H_2O)_6]^{2+}$
   (c) $[Cu(H_2O)_6]^{2+}$ (d) $[Zn(H_2O)_6]^{2+}$

3. Which one of the following complexes is not coloured?
   a) $[Cu(Cl_4)]^{2-}$ b) $[Cd(Cl_4)]^{2-}$ c) $[Fe(Cl_6)]^{4-}$ d) $[Fe(CN)_6]^{4-}$

4. The type of isomerism shown by $[CoCl_2(en)_2]$ is
   a) Ionization b) Geometrical c) Linkage d) Coordination

5. The effective nuclear charge of 2s electron in Beryllium atom is _____
   a) 2.05 b) 1.95 c) 0.70 d) 3.30

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Part – B (2 x 10 = 20 Marks)

6. a. Describe the crystal field splitting of an octahedral complex with neat diagram. Show the CFSE and magnetic moment calculations of both low spin and high spin complexes for the metal ion with d^7 electron configuration. (10 Marks)

(OR)

b. (i) Write a short note on Spectrochemical series. (4 Marks)

(ii) Compare the CFSE and magnetic moment values of $[MnCl_4]^{2-}$ and $[CoCl_4]^{2-}$ and $[CuCl_4]^{2-}$ complexes. [Atomic numbers of Mn, Co and Cu are 25, 27 and 29 respectively]. (6 Marks)

7. a. Discuss the following isomerism shown by the coordination compounds with a suitable example for each. (3+3+4 Marks)

(i) Linkage

(ii) Coordination

(iii) Optical

(OR)

b. (i) Arrange the following complexes in their increasing order of wavelength of light absorbed and explain. (6 Marks)

$[CrF_6]^{3-}$, $[Cr(CN)_6]^{3-}$ and $[Cr(H_2O)_6]^{3+}$

(ii) Calculate the effective nuclear charge experienced by the 3d electron of Zn [Z=30]. (4 Marks)

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DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

SET – I

INTERNAL ASSESSMENT – I (CLA1-T1)

Program : B.Tech
Course Code & Title:21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 13/09/2023
Duration: 08.00 – 08.50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1. In CFT, the strong field ligand such as CO usually produces:
   (a) high spin complexes and small crystal field splitting
   (b) low spin complexes and small crystal field splitting
   (c) low spin complexes and high crystal field splitting
   (d) high spin complexes and high crystal field splitting
2. CFSE for $e^p t_2^q$ configuration of tetrahedral complex is
   (a) $[-0.4p + 0.6q] \Delta_o$ (b) $[-0.27p + 0.18q] \Delta_o$
   (c) $[-0.6p + 0.4q] \Delta_o$ (d) $[-0.5p + 0.5q] \Delta_o$
3. How many unpaired electrons are present in a strong field Mn(II) octahedral complex?
   (a) 0 (b) 1 (c) 2 (d) 4
4. The effective nuclear charge realized by a 1s electron of Helium atom is ……
   (a) 0.7 (b) 0.3 (c) 1.65 (d) 1.7
5. The spin only magnetic moment of $K_3[Fe(C_2O_4)_3]$ is …….
   (a) ~4.9 (b) ~5.9 (c) ~1.73 (d) ~2.83

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Part – B (2 x 10 = 20 Marks)

6. a. i Write a short note on spectrochemical series? Explain with a suitable example, the influence of ligands in the formation of high and low spin complexes. (6 Marks)

ii. $[Ti(H_2O)_6]^{3+}$ is coloured, while $[Sc(H_2O)_6]^{3+}$ is colourless – Explain. (4 marks)

(OR)

b. i. Give proper justification for the statement "Only high spin tetrahedral complexes are known". (6 Marks)

ii. A tetrahedral complex absorbs at 545 nm. What is the respective octahedral crystal field splitting (Δ_o)? (4 Marks)

7 a. i Predict the possible type of isomerism that can exist for the octahedral complex $CrCl_3.6H_2O$. Write down the isomers and elucidate how the isomers can be differentiated? (6 Marks)

ii. Calculate Z* for a 4s electron in Zinc (Z of Zinc is 30) (4 Marks)

(OR)

b. i. $[Co(NH_3)_6]^{3-}$ is diamagnetic and orange yellow. $[CoF_6]^{3-}$ on the other hand is paramagnetic and blue. Justify. (5 Marks)

ii. Explain the salient features and the limitations of CFT. (5 Marks)

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